Nanoscale Device Engineering (EC 502)

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Syllabus

Unit – I L =3, T = 0, P = 0, C = 3

- Introduction to Nanoscience and Nanotechnologies

Unit – 2

- Fabrication Technology & SPICE Device Models

  (VLSI Processes: Introduction, Twin-Well CMOS Process, Integrated Devices, MOSFETs, Resistors, Capacitors, pn Junction Diodes, BiCMOS Process, Lateral pnp Transistor, p-Base and Pinched-Base, Resistors, SiGe BiCMOS Process, VLSI Layout, Beyond 20nm Technology Implications of Technology Scaling: Issues in Deep-Submicron Design: Silicon Area, Scaling Implications, Velocity Saturation, Subthreshold Conduction, PVT Variations, Wiring: The Interconnect, SPICE Device Models: The Diode Model, The Zener Diode Model, MOSFET Models, The BJT Model)

Unit – 3

- Applications of Nanoscience and Nanotechnologies

  (Information and Communication Technologies: Integrated Circuits, Data Storage, Photonics, Displays, Information storage devices, Wireless sensing and communication)

SPICE Models

SPICE Device Models

- A device SPICE (**Simulation Program with Integrated Circuit Emphasis)** model is a mathematical description of an electronic component's behavior used by SPICE circuit simulators to predict its performance.

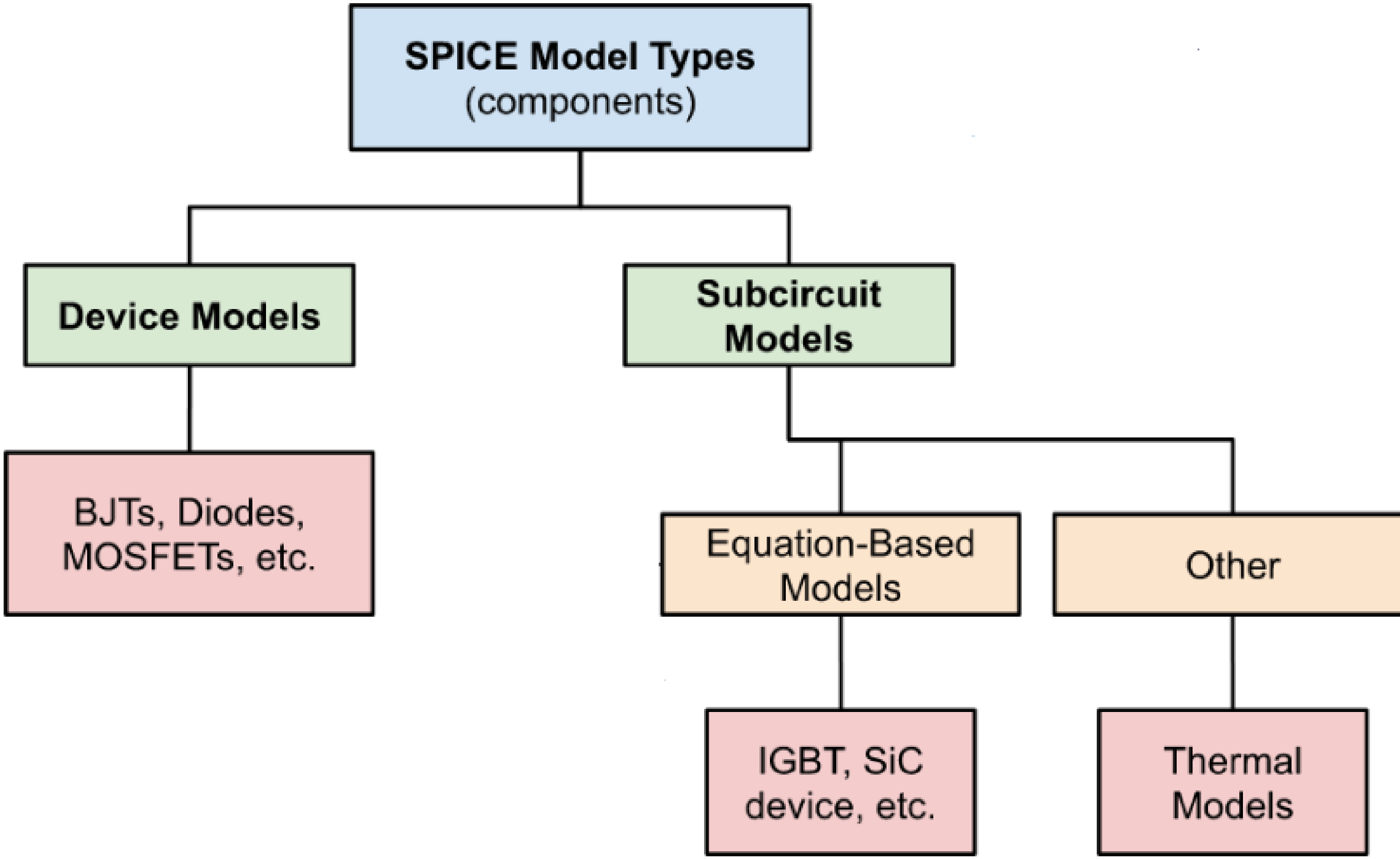
- SPICE was developed at the Electronics Research Laboratory of the University of California, Barkeley by Laurence Nagel in 1973.

- SPICE simulators use a set of mathematical equations to represent a device's behavior, such as its forward current and capacitance for a diode.

- **Example**

- In a diode SPICE model, the current (I_F) is described by an equation that includes the saturation current (I_S) and the diode's voltage (V_d).

SPICE Device Models



SPICE Device Models

SPICE includes following analysis:

- DC analysis (nonlinear quiescent point calculation)
- AC analysis (linear small-signal frequency domain analysis)
- DC transfer curve analysis (a sequence of nonlinear operating points calculated while sweeping an input voltage or current, or a circuit parameter)
- Noise analysis (a small signal analysis done using an adjoint matrix technique which sums uncorrelated noise currents at a chosen output point)
- Transfer function analysis (a small-signal input/output gain and impedance calculation)
- Transient analysis (time-domain large-signal solution of nonlinear differential algebraic equations)

Types of MOSFET SPICE Models

Model Name	Generation / Use	Key Features
Level 1	Basic (long-channel)	Simplified square-law model
Level 2	Improved long-channel	Includes channel-length modulation, mobility variation
Level 3	Semi-empirical	Better short-channel modeling and mobility degradation
BSIM1/2/3/4	Modern CMOS	Accurate short-channel model used in industry
BSIM6 / BSIM-BULK	Advanced bulk MOSFET	Supports deep-submicron, high-K metal gate tech
BSIM-CMG / BSIM-GA / BSIM-IMG	FinFET / GAA / FD-SOI	Include 3D electrostatics and quantum confinement
EKV Model	Analog design	Popular in analog/RF circuit design
HiSIM / PSP Models	Industry-standard for sub-100 nm	Compact models with physics-based parameters

SPICE Device Models

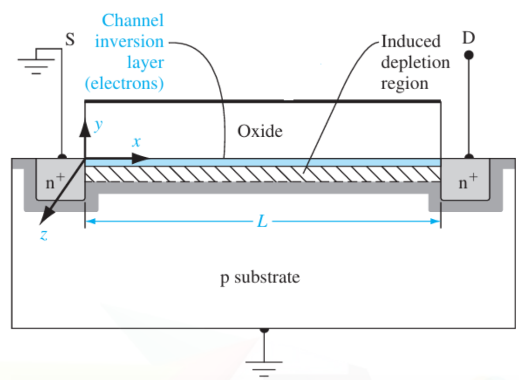
Model Selection in SPICE

Application	Recommended Model
Basic learning / hand analysis	Level 1 to Level 3
General CMOS circuit design	BSIM3 or BSIM4
Deep-submicron (<90 nm)	BSIM6 / PSP
FinFET / GAA devices	BSIM-CMG / BSIM-GA
Analog design focus	EKV / PSP

e.g. .MODEL NMOS NMOS (LEVEL=3 VTO=0.7 KP=50e-6 GAMMA=0.5 PHI=0.6 LAMBDA=0.02 TOX=10n)

Basic SPICE Level-1 MOSFET Model

- The SPICE Level-1 MOSFET Model, also called the Shichman–Hodges model, is a long-channel, square-law model used for MOSFET behavior under the gradual channel approximation.



Step-1:

Assumptions:

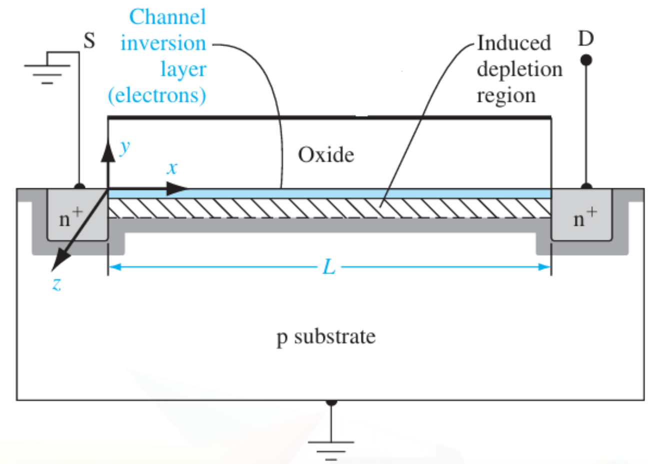
- Long-channel MOSFET
- Constant mobility (μ)
- Gradual-channel approximation ($\partial Ex \ll \partial Ey$) (This approximation means that Ex is a constant.)
- Strong inversion region
- Simple channel-length modulation

Basic SPICE Level-1 MOSFET Model

Step-2: Gradual Channel Approximation

At position x along the channel, the gate-to-channel overdrive voltage is:

$V_{ov}(x) = V_{GS} - V_{th} - V(x)$



Inversion charge per unit area:

$Q_i(x) = - C_{ox} (V_{GS} - V_{th} - V(x))$

$J = \sigma.E = e.\ \mu.\ n_e\ .E = \mu.\ Q_i(x).\ E$

Drain current (drift current):

$I_D = W * \mu * C_{ox} * (V_{GS} - V_{th} - V(x)) * dV/dx$

Basic SPICE Level-1 MOSFET Model

Step-3: Integration for I_D

Integrating from source ($x=0;V=0$) to drain ($x=L;V=V_{DS}$):

$$\Rightarrow \qquad \int_0^L dx = \int_0^{V_{DS}} \frac{W\mu C_{ox}\left(V_{GS} - V_{th} - V\right)}{I_D}\, dV.$$

$$L = \frac{W\mu C_{ox}}{I_D} \int_0^{V_{DS}} \left(V_{GS} - V_{th} - V\right) dV.$$

$$I_D = \mu\, C_{ox}\, (W/L)\, [(V_{GS} - V_{th})V_{DS} - \tfrac{1}{2}V_{DS}^{2}] \quad \text{-------} \; (i)$$

- This expression is valid for the **linear region** ($V_{DS} < V_{GS} - V_{th}$).

Basic SPICE Level-1 MOSFET Model

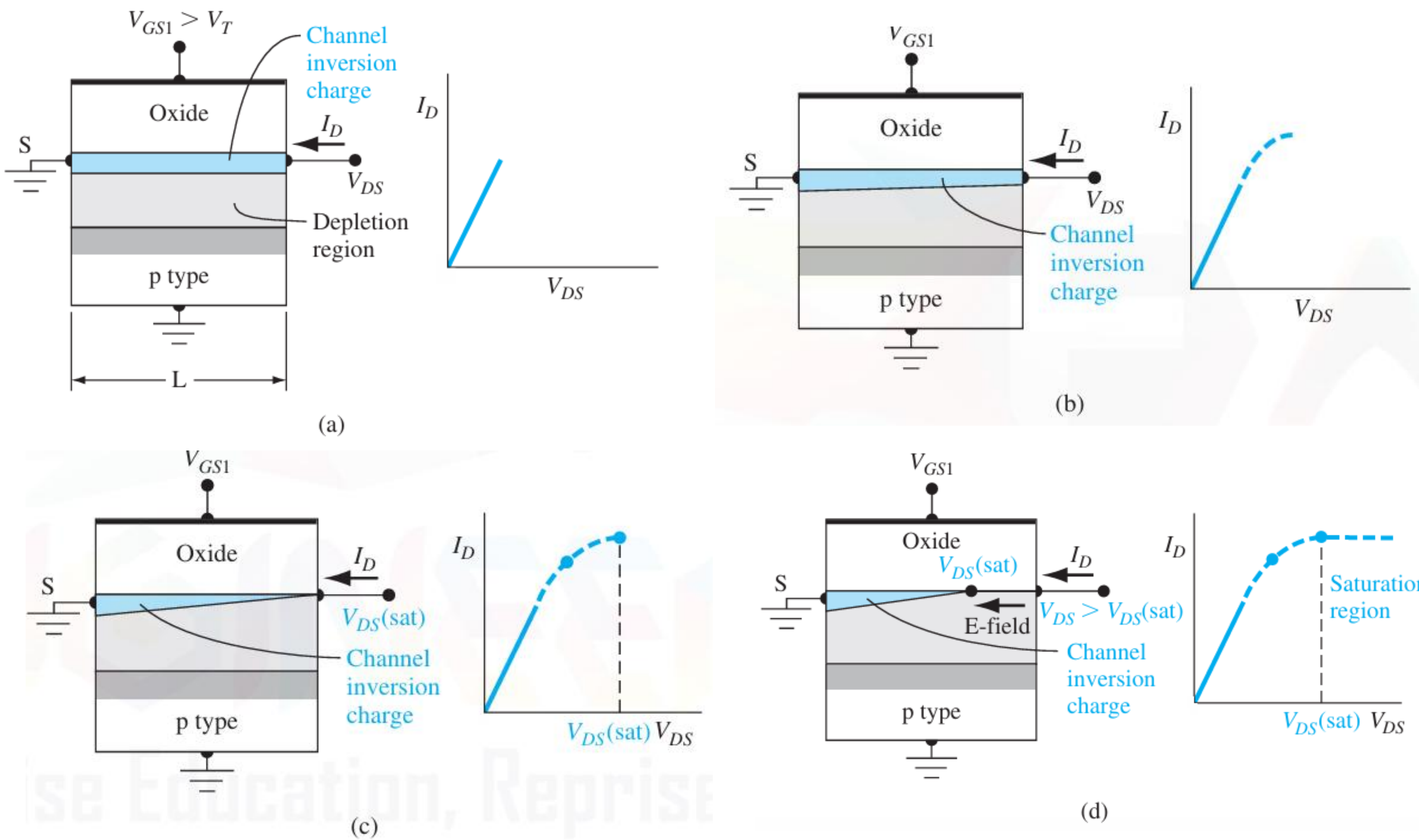


Fig. Cross section and I_D versus V_{DS} curve for (a) small V_{DS} (b) larger V_{DS} (c) $V_{DS} = V_{DS}$(sat), and (d) $V_{DS} > V_{DS}$(sat).

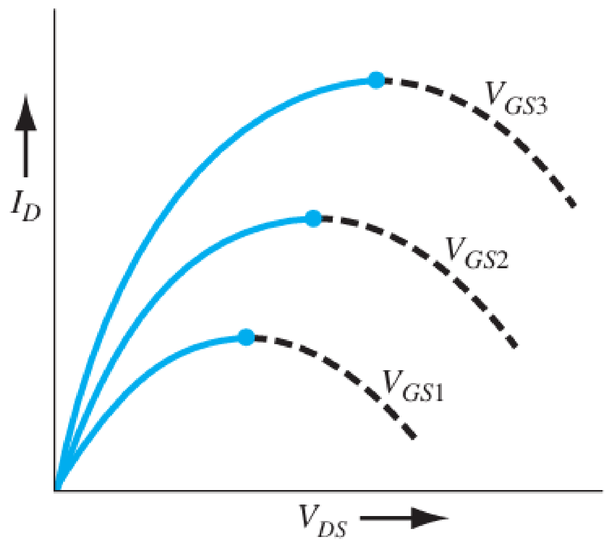
Basic SPICE Level-1 MOSFET Model

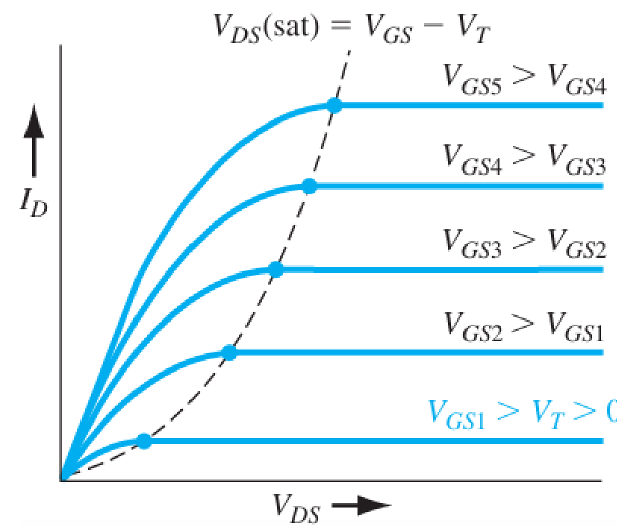
- As the drain voltage increases, the voltage drop across the oxide near the drain terminal decreases, which means that the induced inversion charge density near the drain also decreases.
- When V_{DS} increases to the point where the potential drop across the oxide at the drain terminal is equal to V_T, the induced inversion charge density is zero at the drain terminal.
- When V_{DS} becomes larger than V_{DS}sat value, the point in the channel at which the inversion charge is zero moves towards the source terminal.
- In this case, the electrons in the space charge region are swept by the E-field to the drain contact.

Basic SPICE Level-1 MOSFET Model

$$I_D = \mu\, C_{ox}\, (W/L)\, [(V_{GS} - V_{th})V_{DS} - 1/2 V_{DS}^{\ 2}] \quad \text{------- (i)}$$

- The plot for equation (i) is shown in figure below.





- We can find the V_{DS} value at peak current ($V_{DS} = V_{GS} - V_{th}$) by $\delta I_D/\delta V_{DS}=0$
- Thus, above expression is valid for the **linear region** ($V_{DS} < V_{GS} - V_{th}$).

Basic SPICE Level-1 MOSFET Model

Step-4: Saturation (Active) Region

- When $V_{DS} \geq V_{GS}$ - V_{th}, the channel pinches off at the drain end.
- Set $V_{DS} = V_{GS}$ - V_{th} in the linear-region expression:
- $I_D(sat) = ½ \mu C_{ox} (W/L) (V_{GS} – V_{th})^2$
- This gives the **square-law dependence** of current on gate overdrive.

Basic SPICE Level-1 MOSFET Model

Step-5: Body Effect & Channel-Length Modulation

- Body effect (threshold variation):
- If source-bulk bias $V_{SB} \neq 0$, threshold voltage increases:

$$V_{th} = V_{th0} + \gamma(\sqrt{2\phi_f + V_{SB}} - \sqrt{2\phi_f})$$

- where ϒ is the body-effect coefficient and ϕ_f is the fermi potential.
- Channel length modulation (λ):
- Real transistors show a small dependence of I_D on V_{DS} even in saturation due to effective channel length shortening.
- $I_D = I_D(sat)\,[1 + \lambda V_{DS}]$
- where λ is the **channel-length modulation** parameter (units V^{-1}).
- λ introduces a finite slope in the I_D–V_{DS} curve even in saturation.

Basic SPICE Level-1 MOSFET Model

Step-6: Complete Level-1 Model Summary

Linear region:

- $I_D = \mu C_{ox} (W/L) [(V_{GS} - V_{th})V_{DS} - ½V_{DS}^2]$

Saturation region:

- $I_D = ½ \mu C_{ox} (W/L) (V_{GS} - V_{th})^2 (1 + \lambda V_{DS})$

- Applicable to: Long-channel MOSFETs (pre-submicron era)
- Used in: SPICE Level-1 model for circuit simulation.

Basic SPICE Level-1 MOSFET Model

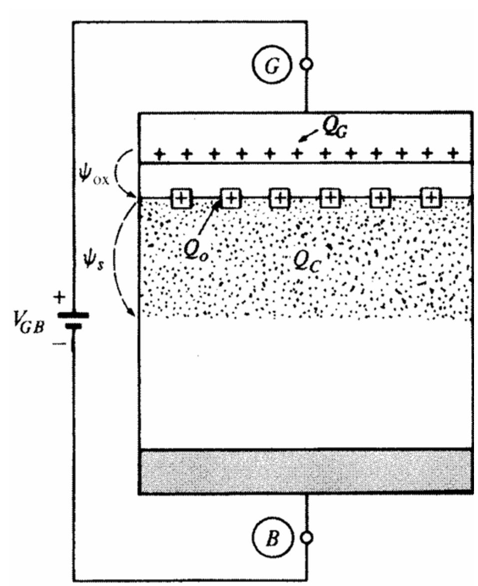
Step-7: Limitations of Level-1 Model

- Ignores short-channel and velocity saturation effects
- Assumes constant mobility (real μ degrades with vertical field)
- Not accurate for modern submicron CMOS
- Replaced by Level-2 and BSIM models in advanced SPICE versions

Basic SPICE Level-1 MOSFET Model

Potential Balance and Charge balance:

- Consider a MOS structure with P-type body. An arbitrary value of V_{GB} cause the charges to appear in the semiconductor.
- We define the surface potential Ψ_S as the total potential drop across the region defined from the surface to a point in bulk.



Basic SPICE Level-1 MOSFET Model

Potential Balance and Charge balance:

- Four types of potential drops are encountered in the loop:

1. The voltage of external source V_{GB}
2. The potential drop across the oxide Ψ_{ox}
3. The surface potential Ψ_S
4. Sum of contact potentials Φ_{MS}

Going around the loop, we can write

$$V_{GB} = \Psi_{ox} + \Psi_S + \Phi_{MS} \quad \text{---------------} \text{ (i)}$$

In equation (i) Φ_{MS} is constant so any changes in VGB will be balanced by Ψ_{ox} and Ψ_S .

$$\Delta V_{GB} = \Delta \Psi_{ox} + \Delta \Psi_S$$

Basic SPICE Level-1 MOSFET Model

Potential Balance and Charge balance:

Now consider the charges in the system. We encounter three types of charges.

1. The Charge on gate Q_G
2. The effective interface charge Q_0
3. The charge in the semiconductor under the oxide Q_C

For charge neutrality:

$Q_G + Q_0 + Q_C = 0$ --------------- (ii)

In terms of charge per unit area

$Q'_G + Q'_0 + Q'_C = 0$

Oxide interface charge is assumed constant, thus

$\Delta Q'_G + \Delta Q'_C = 0$

Basic SPICE Level-1 MOSFET Model

Effect of gate voltage on Surface condition:

- Depending upon whether V_{GB} is less than equal to or greater than V_{FB}, three cases are possible:

1. Flat Band Condition

   $V_{GB} = V_{FB}$

   $Q'_C = 0$

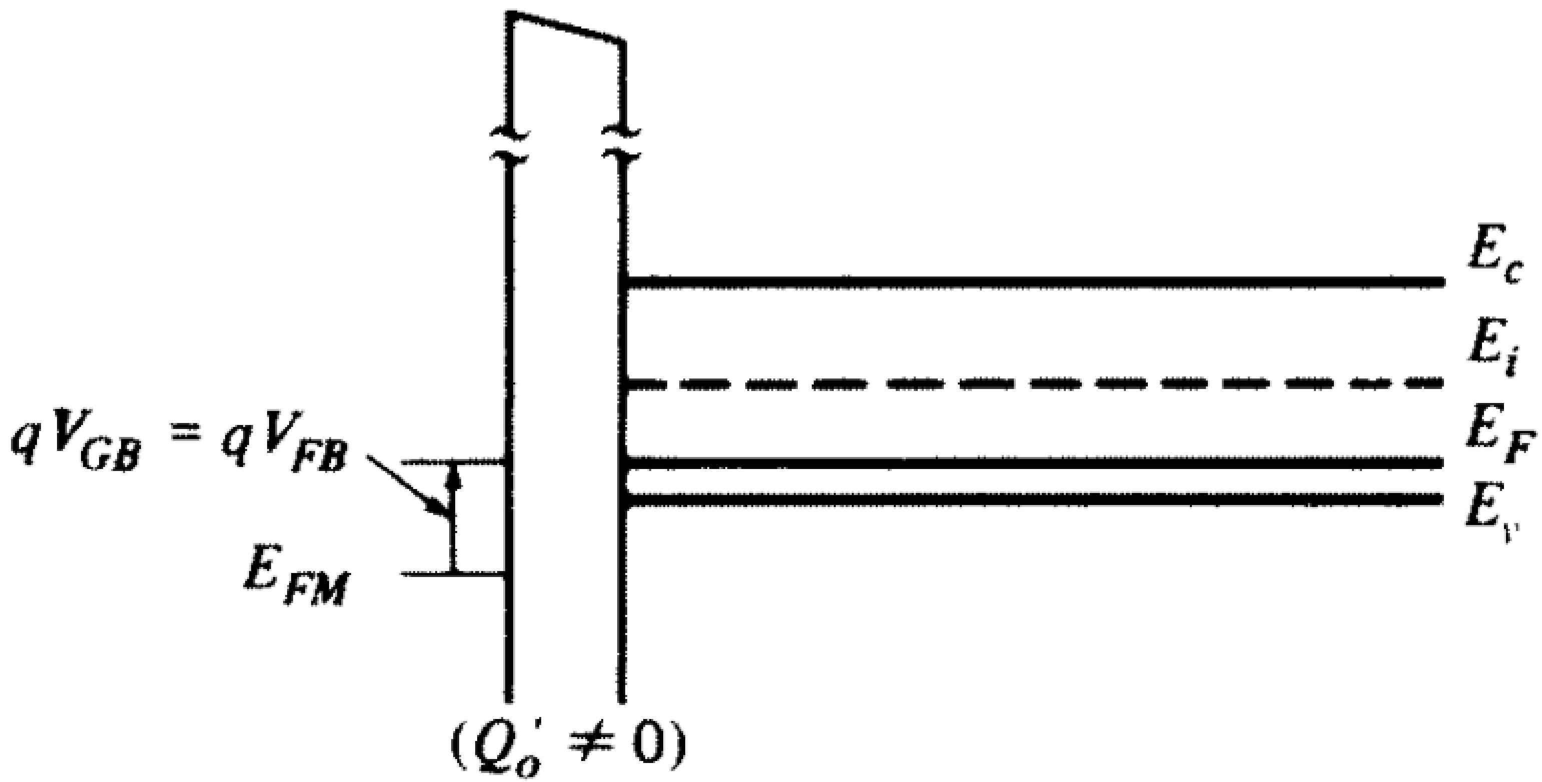
   $\Psi_S = 0$

   This is the external voltage between the gate and substrate terminals to keep the semiconductor everywhere neutral.

- The reason for this name is that with $V_{GB} = V_{FB}$, the energy band diagram in the body is flat due to neutral body.

Basic SPICE Level-1 MOSFET Model

Effect of gate voltage on Surface condition:



Basic SPICE Level-1 MOSFET Model

Effect of gate voltage on Surface condition:

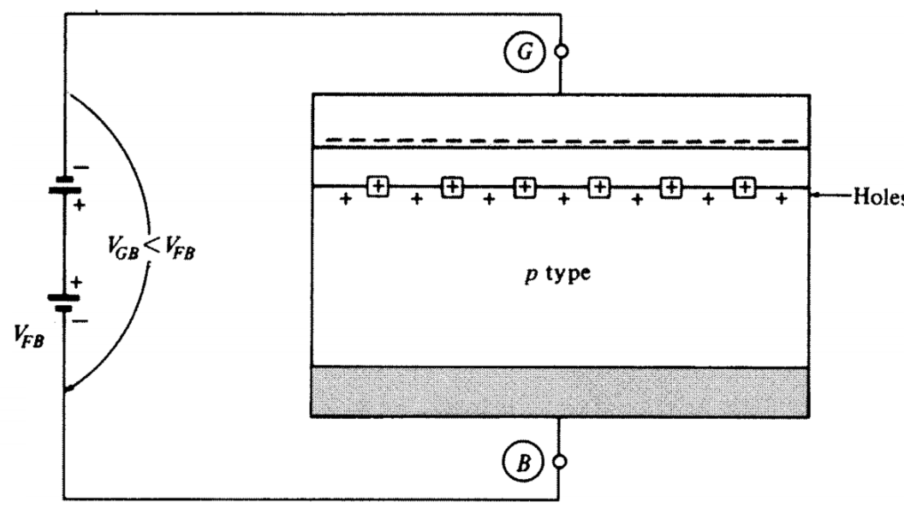
2. Accumulation Condition

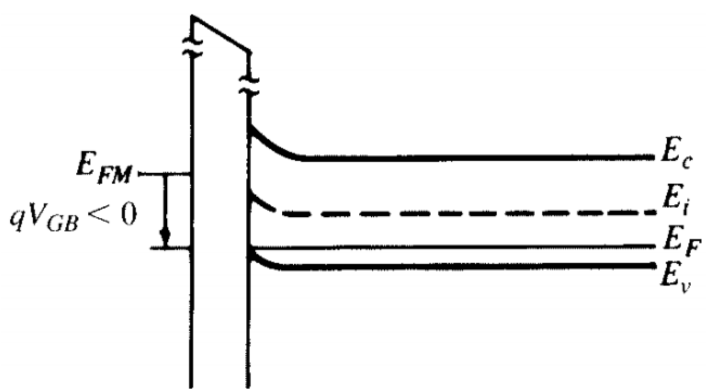
In this condition V_{GB} decreases below V_{FB} . Thus holes will accumulate at the surface to provide a new positive charge.

$V_{GB} < V_{FB}$

$Q'_C > 0$

$\Psi_S < 0$





Basic SPICE Level-1 MOSFET Model

Effect of gate voltage on Surface condition:

3. Depletion and Inversion Condition

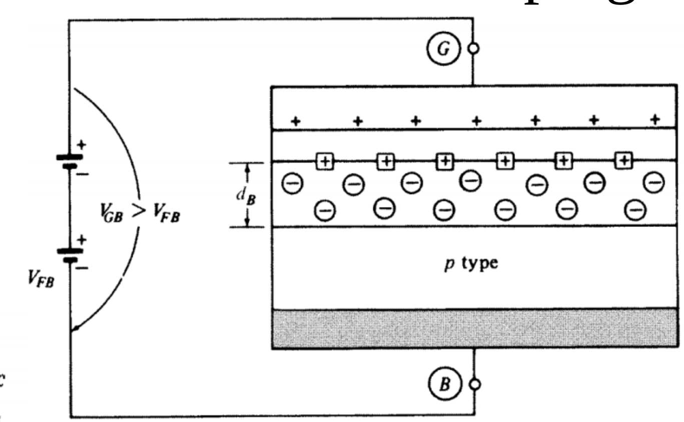
In this condition V_{GB} increases slightly above V_{FB} . Thus holes are removed from the surface leaving negatively charged ions behind.

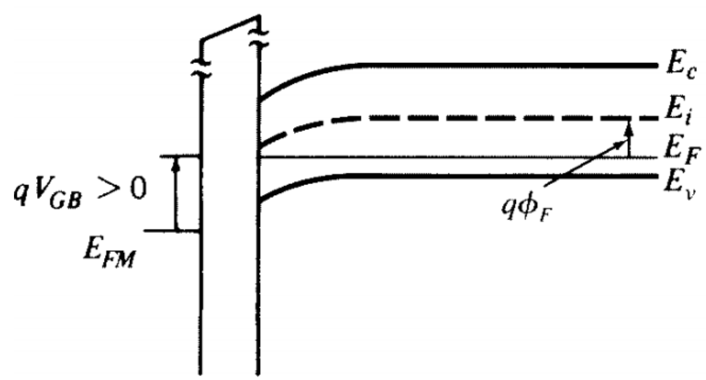
The hole density will decrease below the doping concentration N_A .

$V_{GB} < V_{FB}$

$Q'_C < 0$

$\Psi_S > 0$





Basic SPICE Level-1 MOSFET Model

Effect of gate voltage on Surface condition:

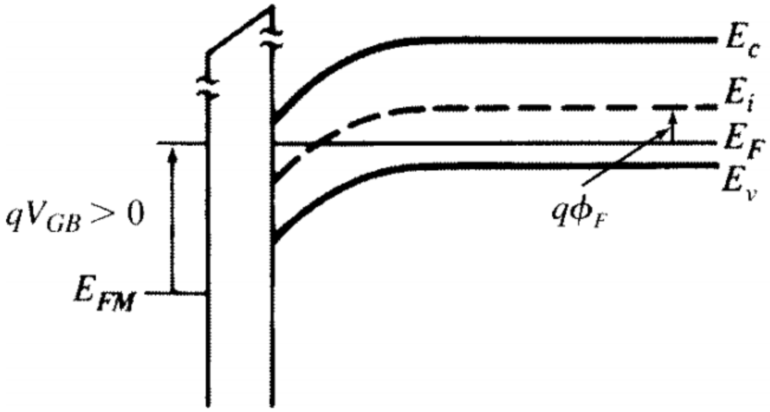
Inversion Condition

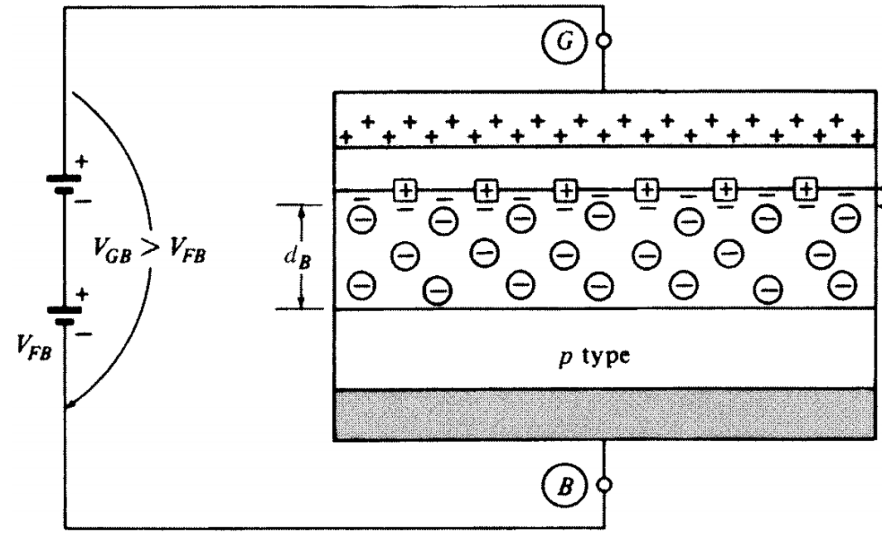
As V_{GB} is increased further above V_{FB} , Ψ_S becomes sufficiently positive to attract significant number of free electrons to the surface.

$V_{GB} < V_{FB}$

$Q'_C < 0$

$\Psi_S > 0$





Basic SPICE Level-1 MOSFET Model

Effect of gate voltage on Surface condition:

3. Depletion and Inversion Condition

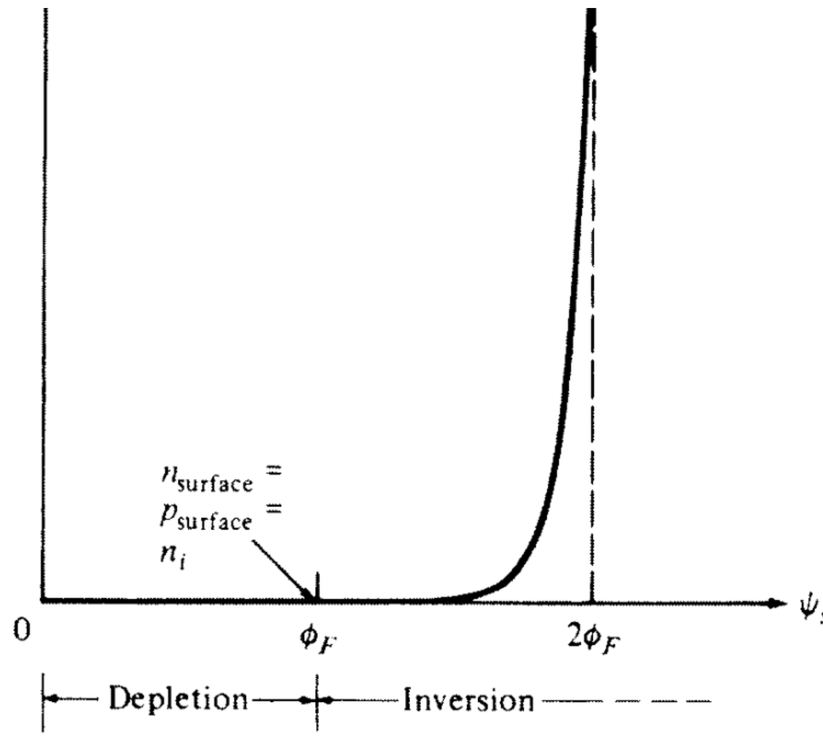


Fig. Electron concentration at the surface vs. surface potential

Basic SPICE Level-1 MOSFET Model

Effect of gate voltage on Surface condition:

Consider a point of ordinate y in the body, and consider Ψ(y) be the potential there with respect to bulk.

The electron concentration at y

$$n(y) = n_o e^{\psi(y)/\phi_t}$$

Where n_0 is the electron concentration in the bulk.

Similarly, hole concentration in the bulk

$$p(y) = p_o e^{-\psi(y)/\phi_t}$$

Where p_0 is the hole concentration in the bulk.

Basic SPICE Level-1 MOSFET Model

Effect of gate voltage on Surface condition:

Total charge density

$$\rho(y) = q[p(y) - N_A + N_D - n(y)]$$

Thus Poisson's equation can be written as

$$\frac{d^2\psi}{dy^2} = -\frac{q}{\epsilon_s}\left[p_o e^{-\psi(y)/\phi_t} - N_A + N_D - n_o e^{\psi(y)/\phi_t}\right]$$

Thanks..!!!